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ORIGINAL PAPER

GTUNet: a GNN and transformer enhanced U-Net for coronary artery segmentation

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Abstract

Accurate segmentation of complex biological structures, particularly coronary arteries, remains a significant challenge due to their intricate topologies and multi-scale morphological variations. In this paper, we propose GTUNet, a novel hybrid framework that synergistically integrates Graph Attention Networks, Transformers, and UNet for robust 3D medical image segmentation. The architecture features dual parallel encoder-decoder branches utilizing 3D ResUNet and Swin Transformer to capture multi-level representations. To encode geometric priors, a GAT-enhanced module is introduced specifically for the CNN branch to refine topological relationships before cross-stream interaction. Furthermore, a Bi-directional Attention Fusion Module is designed to bridge the two streams: it utilizes dual attention mechanisms to inject high-frequency local details from the CNN into the Transformer, while simultaneously transferring long-range dependencies from the Transformer back to the CNN. Extensive evaluations on the ASOCA and ImageCAS datasets demonstrate that GTUNet achieves state-of-the-art performance in Dice scores, ensuring superior topological continuity. Notably, the model significantly outperforms existing methods in ASSD, highlighting its precision in delineating vascular boundaries.

Keywords Coronary artery segmentation · Graph attention networks · Swin transformer · Feature fusion and enhancement

1 Introduction

Coronary Artery Disease (CAD) remains a global health priority, with Coronary Computed Tomography Angiography (CCTA) serving as the non-invasive gold standard. High-fidelity 3D coronary segmentation is a prerequisite for anatomical modeling and functional indices like Fractional Flow Reserve (FFR-CT). However, automated segmentation is hindered by the coronary tree's structural sparsity, high anisotropy, and complex branching. These challenges are exacerbated by cardiac motion and calcification blooming, often causing "disconnection artifacts" and the omission of distal branches, which compromises downstream quantitative analysis.

While U-Net-based CNNs excel at local representation, their inductive biases–locality and translation invariance–limit their perception of global hierarchies, often yielding fragmented topologies. Recent hybrid architectures integrate Vision Transformers (ViTs) or Graph Neural Networks (GNNs) to transcend these limits. However, ViTs often overlook strict topological constraints, while GNN-CNN hybrids frequently lack the multi-scale global context necessary for whole-tree integrity. Consequently, synergizing local cues, global semantic dependencies, and non-Euclidean topological priors remains elusive.

To address these limitations, we propose GTUNet, a hybrid framework for topology-aware coronary segmentation. GTUNet employs a dual-stream architecture featuring a 3D ResUNet and a parallel Swin Transformer. A Graph Attention Network (GAT) module is integrated into the CNN branch to embed non-Euclidean structural priors. To synergize these features, we design a Bidirectional Attention Fusion Module (BAFM) that facilitates reciprocal information exchange: high-frequency textures refine the Transformer branch, while long-range hierarchies stabilize the CNN's perception. This integration ensures GTUNet captures both intricate bifurcations and global continuity. The primary contributions of this work are as follows:

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- We introduce GTUNet, a novel hybrid architecture that synergistically integrates 3D-CNNs, Swin Transformers, and GATs. This framework effectively bridges the gap between local texture perception, long-range global context, and non-Euclidean topological priors to ensure superior vascular continuity.
- We propose a GAT-driven enhancement module to embed structural connectivity into CNN-based local features, alongside a Bidirectional Attention Fusion Module (BAFM) that facilitates reciprocal information exchange between heterogeneous streams.
- We conduct extensive evaluations on the public ASOCA and ImageCAS datasets, demonstrating that GTUNet achieves new state-of-the-art benchmarks. Our method exhibits exceptional performance in maintaining clinical-grade topological fidelity, effectively mitigating fragmentation in distal and sparse vascular branches.

2 Related work

The paradigm of medical image segmentation has been fundamentally reshaped by CNNs, with U-Net [1] as the cornerstone. Innovations like 3D-UNet [2] and Dense-UNet [3] extended this to volumetric modeling and feature reuse, while nnU-Net [4] established state-of-the-art benchmarks via automated hyperparameter adaptation. To mitigate the receptive field bottleneck, research has pivoted toward multi-scale context aggregation [5] and non-local attention. For vascular structures, Dynamic Snake Convolution [6] and HarmonySeg [7] address elongated morphology and boundary ambiguities. Despite these advancements, CNNs struggle to maintain global topological consistency in sparse, intricately branched vascular trees.

Transformers, spearheaded by ViT [8], utilize self-attention to capture long-range dependencies, leading to hybrid frameworks like TransUNet [9]. For 3D tasks, models such as UNETR [10], Swin-Unet [11], and nnFormer [12] leverage shifted windows or deformable attention [13] to balance modeling capacity with computational complexity. Nevertheless, the lack of inductive biases in Transformers often results in sub-optimal performance on fine-grained vascular boundaries compared to CNNs.

Vessel segmentation necessitates strict topological fidelity, which standard overlap-based losses like Dice often fail to ensure. Early methodologies introduced geometric constraints via the Deep Distance Transform [14] and clDice [15]. Recent advances have shifted toward sophisticated, domain-specific topological priors; for instance, DeformCL [16] used deformable centerlines to mitigate structural fractures inherent in discrete masks. Parallelly, Graph Neural Networks (GNNs) have emerged as a potent paradigm for non-Euclidean structural modeling [17]. However, dual-branch attempts like ViG3D-UNet [18] often struggle with "cold-start" initialization issues when attempting to construct graph representations directly from unrefined raw images. To alleviate such bottlenecks without explicit graph construction, recent alternative paradigms explore embedding geometric and structural priors directly into network architectures. Along this line, architectural mechanisms have been proposed to enforce localized connectivity and interactivity [19], sequentially integrate hierarchical edge and morphology features [20], or incorporate bilateral symmetry as a learnable global scanning bias [21].

GTUNet addresses these gaps by emphasizing early-stage synergy between convolutional features and graph-based reasoning. Unlike prior architectures that treat GNNs as auxiliary post-processors, GTUNet integrates a GAT-augmented branch to perform topological enhancement directly on semantic feature maps. This ensures that CNN-derived descriptors are inherently imbued with global topological priors, robustly capturing intricate connectivity while overcoming initialization bottlenecks common in existing frameworks.

3 Method

3.1 Overview of the GTUNet

The proposed GTUNet is a topology-aware hybrid framework designed to address topological discontinuity and structural sparsity in 3D coronary artery segmentation. As illustrated in Fig. 1, the architecture follows a dual-stream paradigm composed of a 3D ResUNet branch and a parallel Swin Transformer branch, which capture local texture patterns and long-range contextual dependencies, respectively. The main design lies in the integration of a GAT Enhancement Module and a Bidirectional Attention Fusion Module (BAFM). Specifically, the GAT module is embedded within the CNN branch to introduce non-Euclidean structural priors before cross-stream interaction, thereby strengthening the representation of vascular connectivity. Subsequently, BAFM enables reciprocal feature exchange between the two branches, aligning the high-frequency local details from CNNs with the global semantic context from Transformers. Through this complementary interaction, GTUNet can better preserve fine-grained boundary details while maintaining long-range structural continuity in complex coronary artery trees.

3.2 Parallel double-branch structure

To capture multi-scale feature representations, GTUNet leverages two specialized encoder-decoder pathways. The CNN branch is built upon a 3D ResUNet backbone, uti-

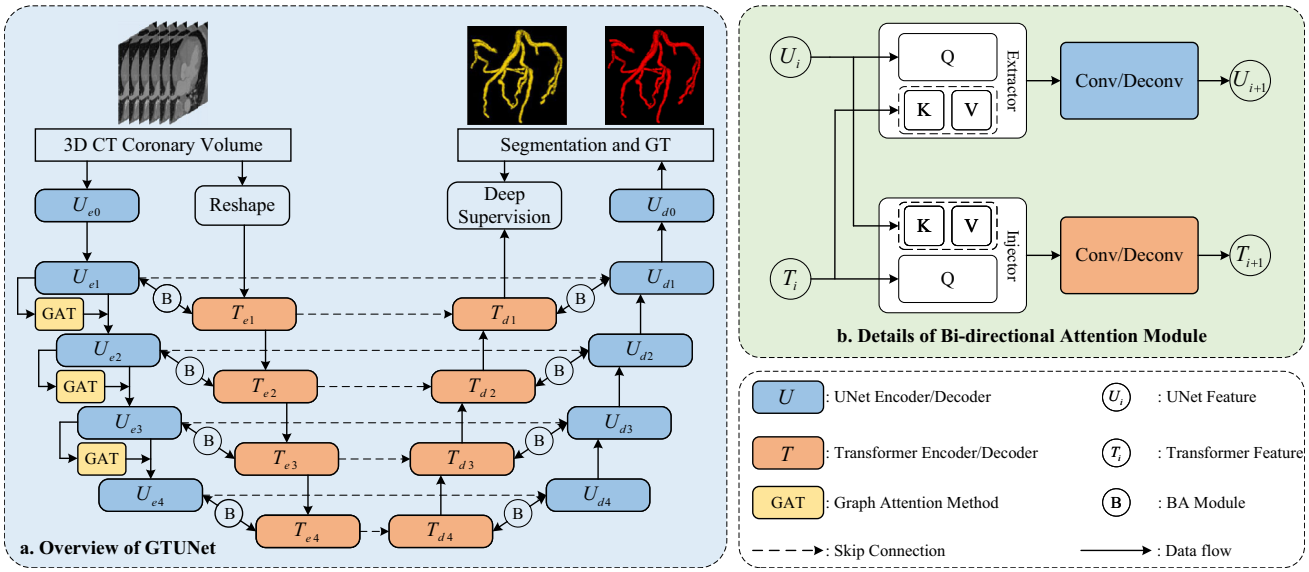


Fig. 1 Architecture of GTUNet. **a** Overall framework of GTUNet, which consists of a 3D ResUNet branch and a Swin Transformer branch for parallel feature extraction. The GAT module is embedded in the CNN branch to enhance structural representation, and BAFM is used for cross-stream feature interaction. **b** Details of the Bidirectional Attention Fusion Module, where CNN and Transformer features are mutually recalibrated through attention-guided information exchange

lizing residual blocks to alleviate gradient vanishing while extracting robust local intensity cues and textural patterns. Simultaneously, the Transformer branch employs a Swin Transformer architecture, which utilizes a shifted-window multi-head self-attention (SW-MSA) mechanism. This hierarchical design allows the Transformer to model global anatomical hierarchies with a linear computational complexity relative to the image size. By operating in parallel, these two branches provide complementary perspectives: the CNN focuses on voxel-level discriminative features, while the Transformer encapsulates the volumetric spatial dependencies necessary for understanding the complex branching of the coronary tree.

3.3 GAT enhancement module

Recognizing the complex curvilinear morphology and topological constraints of coronary arteries, we introduce a Graph Attention Networks (GAT) Enhancement Module to reinforce the CNN branch (Fig. 2). This module transforms Euclidean feature maps into a graph-structured representation to explicitly model vascular skeletal connectivity.

3.3.1 Graph construction

While GNNs excel at topological preservation, a gap exists between their non-Euclidean nature and the structured Euclidean grids of medical imaging. To bridge this, we adopt a K-Nearest Neighbor (KNN) graph construction–preferring it over superpixel clustering–to more faithfully encapsulate the curvilinear morphology and skeletal continuity of coronary arteries.

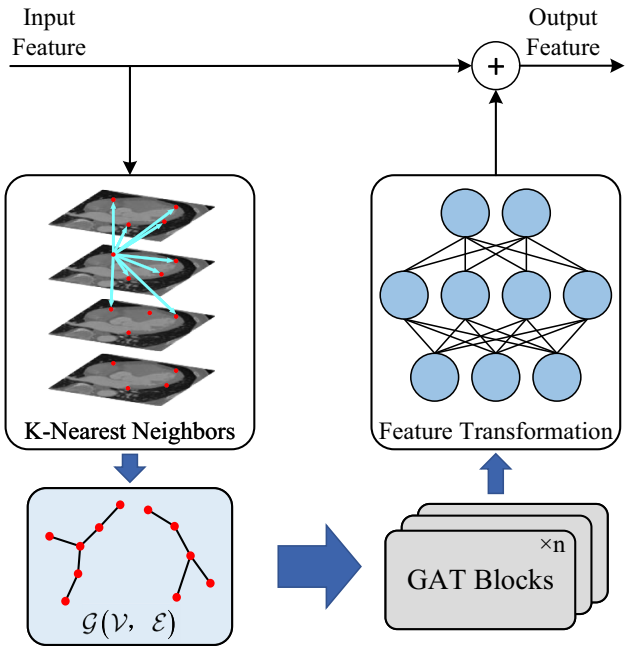


Fig. 2 Structure of the GAT enhancement module, including KNN graph construction, stacked GAT blocks, and feature transformation

To optimize the trade-off between computational efficiency and topological awareness, the GAT module is integrated after the second convolutional stage. This ensures the

graph is derived from feature maps with sufficient semantic depth while maintaining manageable spatial resolution.

For each node i at spatial coordinate $p_i \in \mathbb{R}^3$, the local neighborhood $\mathcal{N}_i$ is established by minimizing the L_2 distance:

$$d_{ij} = |p_i - p_j|2 = \sqrt{\sum k \in x, y, z(i_k - j_k)^2}. \quad (1)$$

The adjacency neighborhood $\mathcal{N}_i$ and the resulting edge set $\mathcal{E}$ are formally defined as:

$$\begin{aligned} \mathcal{N}_i &= \{j \in \mathcal{V} \setminus \{i\} \mid d_{ij} \leq \delta_i^{(K)}\}, \\ \mathcal{E} &= \{(i, j) \mid j \in \mathcal{N}_i, \forall i \in \mathcal{V}\}. \end{aligned} \quad (2)$$

where $\delta_i^{(K)}$ denotes the distance to the K-th nearest neighbor. This sparse connectivity constrains the search space to the K closest spatial neighbors, filtering non-vascular noise and reducing computational complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(NK)$.

3.3.2 Graph processing and feature transformation

The constructed graph $\mathcal{G}$ is processed through a stack of L GAT blocks to facilitate deep topological reasoning [22]. Unlike standard GCNs, GAT adaptively assigns weights to different neighbors, which is crucial for modeling the irregular branching of coronary arteries. Within each block, a shared linear transformation parameterized by $\mathbf{W} \in \mathbb{R}^{F' \times F}$ maps every node feature $\vec{h}_i$ into a high-dimensional space. The raw alignment score e_{ij} between a center node i and its neighbor $j \in \mathcal{N}_i \cup \{i\}$ is computed via a parameterized self-attention mechanism:

$$e_{ij} = \text{LeakyReLU}\left(\vec{a}^T [\mathbf{W}\vec{h}_i, \|, \mathbf{W}\vec{h}_j]\right), \quad (3)$$

where $\vec{a} \in \mathbb{R}^{2F'}$ is a learnable weight vector and $\|$ denotes concatenation.

To ensure cross-neighborhood comparability, scores are normalized using the softmax function:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}i \cup i} \exp(eik)}. \quad (4)$$

To stabilize learning and capture diverse topological patterns, the mechanism is extended to K independent attention heads. The aggregated output feature $\vec{h}'_i$ is obtained by concatenating the multi-head representations, followed by a non-linear activation σ:

$$\vec{h}'i = \sigma\left(\Big\|_{k=1}^{K} \sum j \in \mathcal{N}i \cup i\alpha ij^k \mathbf{W}^k \vec{h}_j\right), \quad (5)$$

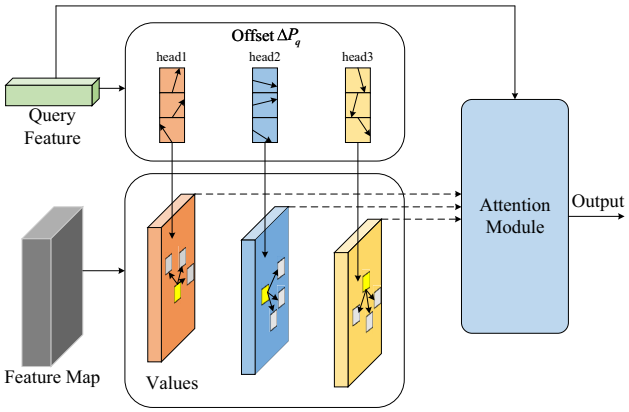


Fig. 3 Structure of the bidirectional attention fusion module based on deformable attention

where α_{ij}^k and $\mathbf{W}^k$ denote the coefficients and weight matrix of the k-th head.

Through the dynamic calculation of α_{ij}^k, this multi-head mechanism acts as an adaptive topological filter that regularizes global structural fidelity. By modulating feature propagation based on semantic alignment rather than spatial proximity, the normalized coefficients prevent erroneous message leakage across adjacent but separate vascular pathways. Crucially, this gated information flow reinforces branch continuity; the attention heads inherently favor interconnected vessel segments to bridge local fractures, while stacking L layers enables localized attention fields to propagate over multiple hops (Fig. 3). This multi-hop traversal allows the model to capture distal branch continuity and global hierarchies without explicit graph-wide tracking. Following this aggregation phase, a dual-layer MLP-based feed-forward network (FFN) refines each node, consolidating intra-branch relational dependencies and distilling latent semantic representations to ensure a robust mapping of structural priors.

3.4 Bidirectional attention fusion module

To bridge the semantic divergence between heterogeneous streams, we propose the Bidirectional Attention Fusion Module (BAFM) to facilitate cross-modal information flow. The BAFM executes a reciprocal interaction via two complementary pathways: Local-to-Global Injection, where fine-grained CNN features refine Transformer representations with precise spatial cues; and Global-to-Local Propagation, where holistic vascular context from the Transformer is back-propagated to restore connectivity in fragmented CNN segments. This synergy ensures that the network effectively reconciles high-frequency local textures with expansive global hierarchies to yield a robust, topology-preserving latent representation.To maintain high efficiency, BAFM utilizes Deformable Attention (DA) rather than vanilla self-

attention, significantly mitigating computational complexity by focusing on a sparse set of key sampling points. Given a query element q with its content feature z_q and a 3D reference point p_q, the Deformable Attention is formulated as:

$$\mathrm{DA}(z_q, p_q, F) = \sum_{m=1}^{M} W_m h_{mq}, \tag{6}$$

where h_{mq} represents the aggregated feature for the m-th attention head:

$$h_{mq} = \sum_{k=1}^{K} A_{mqk} \cdot W'mF(p_q + \Delta pmqk), \tag{7}$$

Here, M and K denote the number of attention heads and sampling points, respectively. Δp_{mqk} is the learnable sampling offset for the k-th point in the m-th head, enabling the model to adaptively conform to irregular vascular morphologies. $A_{mqk} \in [0, 1]$ is the normalized attention weight determined by the query feature z_q, and $F(\cdot)$ represents the feature map sampled via trilinear interpolation.In the CNN-to-Transformer pathway, CNN features serve as the key/value source F while Transformer features provide the query z_q, embedding local textures into the global context. Conversely, during Transformer-to-CNN propagation, the roles are inverted to back-propagate long-range spatial dependencies, which is instrumental in mitigating disconnection artifacts in distal coronary branches.

4 Experiments

4.1 Datasets

To rigorously evaluate the generalization and robustness of GTUNet, we benchmark our method on two diverse datasets. ASOCA: The Automated Segmentation of Coronary Arteries dataset from the MICCAI 2020 Challenge consists of 60 Cardiac CT Angiography (CCTA) scans. ImageCAS [25]: This large-scale benchmark contains 1000 3D CTA scans from patients with coronary artery disease. It captures significant anatomical complexity, including post-revascularization cases.

4.2 Experimental settings

All experiments were implemented in PyTorch using an NVIDIA RTX 4070 Super GPU. We employed the Adam optimizer with an initial learning rate of 1×10^{-4}, decayed by a factor of 10 every 50 epochs over a 400-epoch schedule. To manage memory constraints, we adopted a patch-based strategy with an input size of $112 \times 112 \times 112$ and a batch size of 1. The composite loss function integrated Dice Loss, Cross-Entropy Loss, and Centerline Dice (clDice) to explicitly penalize topological disconnectivity.

4.3 Evaluation metrics

We employed a comprehensive set of metrics to evaluate both the volumetric accuracy and the topological continuity of the segmentation results.

We utilized the Dice Similarity Coefficient (DICE) and the Mean Symmetric Surface Distance (ASSD). The Dice score is defined as:

$$DICE = \frac{2|S \cap G|}{|S| + |G|}, \tag{8}$$

where S denotes the segmentation prediction and G denotes the ground truth. ASSD measures the average distance between the surface of the segmentation S and the ground truth G:

$$\mathrm{ASSD}(S, G) = \frac{\sum_{s \in S_s} d(s, G_s) + \sum_{g \in G_s} d(g, S_s)}{|S_s| + |G_s|}, \tag{9}$$

where S_s and G_s represent the surface voxel sets of S and G, respectively, and $d(\cdot)$ is the Euclidean distance.

Given the tubular nature of vessels, we employed Skeleton Recall (SR) and Skeleton Precision (SP) [15] to assess topological integrity.

$$\mathrm{SR}(S, G) = \frac{|S \cap Q(G)|}{|Q(G)|}, \tag{10}$$

$$\mathrm{SP}(S, G) = \frac{|G \cap Q(S)|}{|Q(S)|}. \tag{11}$$

where Q(*) is the skeletonization function [26] to preserve the vessel topology and connectivity.

Furthermore, we calculated voxel-level Precision, Recall (Sensitivity), and Intersection over Union (IoU) based on the confusion matrix elements.

4.4 Comparison with the reference models

We benchmarked GTUNet against four representative methods spanning CNN-based and Transformer-based approaches, including 3D ResUNet [23], nnU-Net [4], UNETR [10], and CTN [24].

The quantitative results in Table 1 show that GTUNet achieves state-of-the-art performance on both datasets. Notably, its strong competitiveness on the small ASOCA dataset indicates favorable data efficiency and robustness, ensuring stable clinical deployment across centers with varying imaging protocols and scanner modalities (Fig. 4). This

Table 1 Quantitative comparison with four representative segmentation methods on the ASOCA and ImageCAS datasets

Dataset	Method	Dice (%) ↑	ASSD ↓	SR (%) ↑	SP (%) ↑	IoU ↑	Prec(%) ↑	Recall(%) ↑
ASOCA	3D ResUNet [23]	78.62	6.43	82.46	81.35	66.23	82.72	82.23
	nnUNet [4]	83.51	3.86	83.53	82.47	71.72	83.27	83.82
	UNETR [10]	77.97	6.24	82.12	82.04	65.76	80.26	79.92
	CTN [24]	85.21	3.37	86.35	85.27	77.53	85.92	86.13
	GTUNet	**86.85**	**1.12**	**89.15**	**87.34**	**78.92**	**88.13**	**89.26**
ImageCAS	3D ResUNet [23]	77.52	5.87	81.42	80.25	63.35	81.23	81.13
	nnUNet [4]	82.32	3.15	82.36	81.36	70.12	82.93	83.17
	UNETR [10]	77.37	6.41	81.64	81.19	63.21	79.76	79.64
	CTN [24]	84.57	2.47	83.72	83.96	73.74	84.47	85.25
	GTUNet	**85.74**	**1.65**	**85.45**	**84.72**	**75.72**	**85.46**	**86.69**

Bold values indicate the best performance achieved among all compared methods

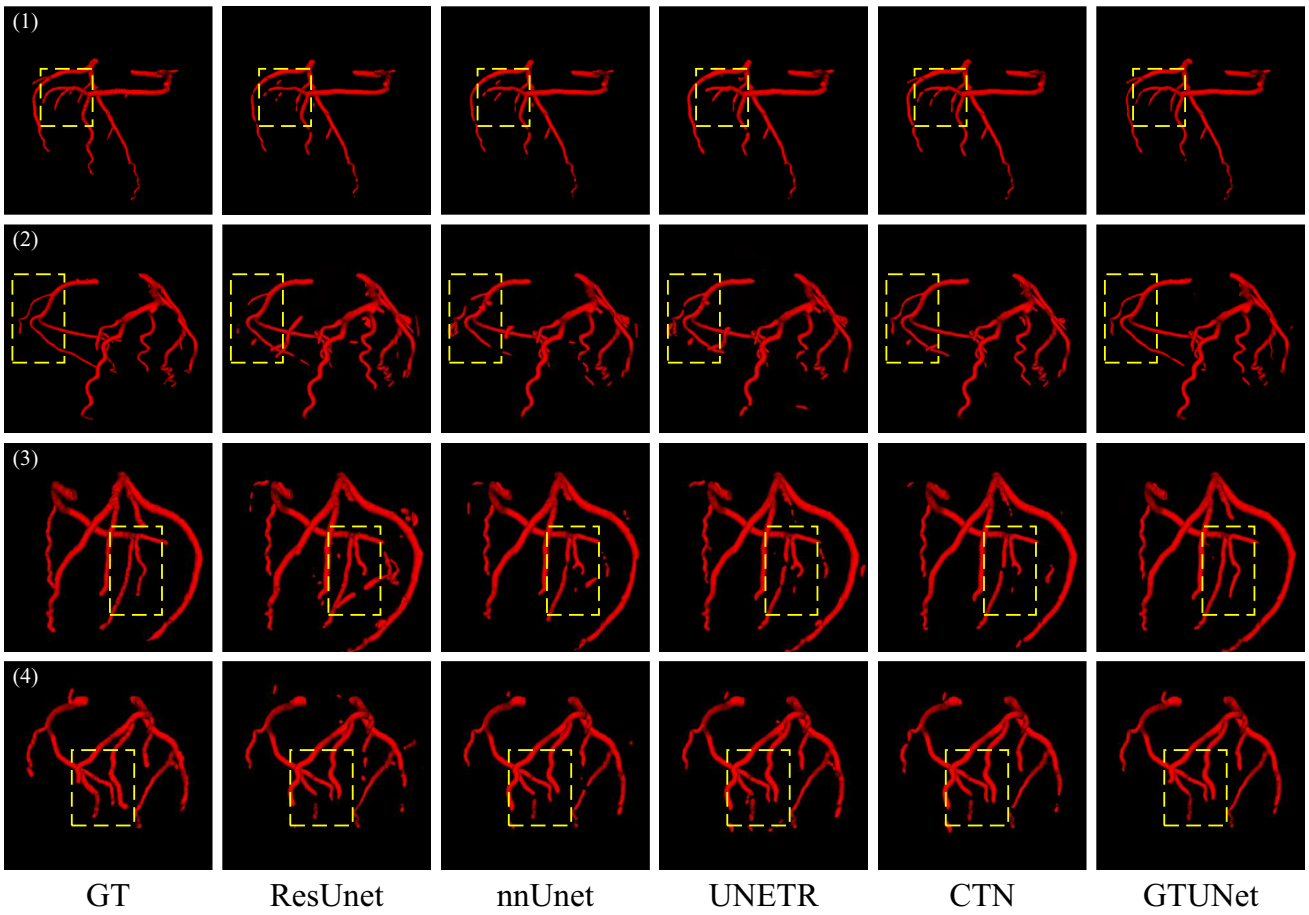


Fig. 4 Qualitative comparison of coronary artery segmentation results on the ASOCA and ImageCAS datasets. Samples (1)–(2) and (3)–(4) are from ASOCA and ImageCAS, respectively. Yellow boxes highlight regions where GTUNet better preserves vascular continuity and fine-grained structures, reducing fragmentation and omission errors compared with baseline models

architectural advantage enables GAT and Transformers to effectively capture intricate spatial topologies from limited samples. For volumetric metrics (Dice and IoU), GTUNet consistently outperforms competing methods, demonstrating precise lumen boundary delineation for accurate stenosis grading. More importantly, its significant advantage in Average Symmetric Surface Distance (ASSD) is clinically vital for coronary CTA-derived fractional flow reserve (FFR-

Table 2 Ablation study of different feature fusion strategies on the ImageCAS dataset

Fusion method	Dice (%)↑	ASSD↓	IoU (%)↑
Addition	85.07	2.24	74.86
Concatenation	85.24	2.17	75.05
BAFM	**85.74**	**1.65**	**75.72**

Bold values indicate the best performance achieved among all compared methods

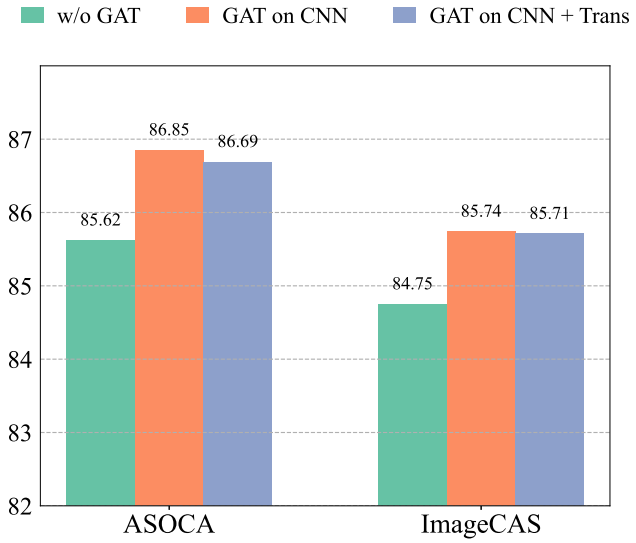


Fig. 5 Ablation analysis of different GAT integration strategies

CT) and high-fidelity luminal modeling; minimizing this surface deviation prevents sub-millimeter errors from causing substantial deviations in hemodynamic simulations and subsequent ischemia misdiagnosis. Furthermore, the clear advantages in Skeleton Recall (SR) and Skeleton Precision (SP) confirm a superior balance between voxel accuracy and global continuity. Specifically, the higher SR minimizes the risk of missing distal branches and fine peripheral lesions, while the high SP maintains centerline continuity, fulfilling critical clinical requirements for automated vessel tracking, curved planar reformation (CPR), and virtual stenting planning without tedious manual corrections.

4.5 Ablation study

4.5.1 Ablation of fusion method

To evaluate the effectiveness of the Bidirectional Attention Fusion Module (BAFM), we compared it against two standard fusion strategies: element-wise addition (Add) and channel-wise concatenation (Concat). As shown in Table 2, BAFM achieves the best performance across all metrics. While Add and Concat rely on static feature aggregation

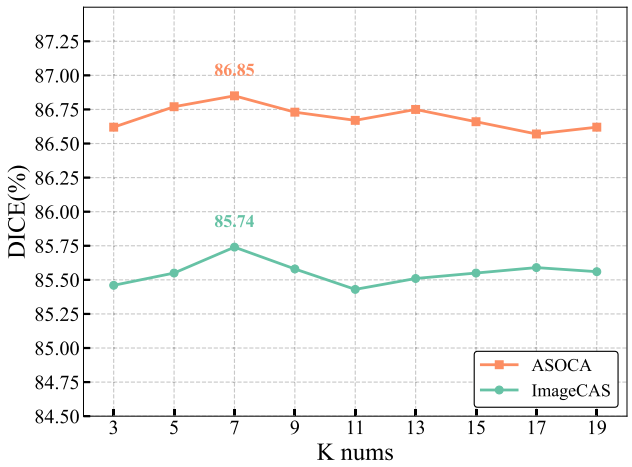


Fig. 6 Sensitivity analysis of the KNN neighborhood size in the GAT module

(linear summation or channel stacking followed by convolution), they fail to model the semantic discrepancy between CNN-based local features and Transformer-based global contexts. In contrast, BAFM adaptively recalibrates features from both streams via bidirectional attention, allowing the CNN branch to incorporate global context while preserving fine-grained local details in the Transformer branch. Consequently, BAFM yields more coherent coronary segmentations, particularly in challenging bifurcation regions and thin distal branches.

4.5.2 Ablation on GAT module

To assess the contribution of the Graph Attention Network (GAT), we first verified its overall effectiveness. As shown in Fig. 5, integrating GAT consistently outperforms the dual-stream baseline, confirming that modeling non-Euclidean structural priors helps preserve the topology of sparse coronary trees. We further examined its positioning by comparing our intra-CNN enhancement against applying GAT to concatenated cross-stream features. Although the latter incorporates richer information, it yields lower accuracy. This suggests that while BAFM already ensures sufficient global interaction, the CNN branch specifically benefits from targeted structural refinement. Moreover, the intra-CNN design avoids high-dimensional graph construction overhead, offering a superior balance between segmentation accuracy and computational efficiency.

4.5.3 Sensitivity analysis of KNN neighborhood size

To optimize the GAT module, we evaluated the neighborhood hyperparameter $k \in \{3, 5, \ldots, 19\}$. As shown in Fig. 6, Dice scores on the ASOCA and ImageCAS datasets peak at $k = 7$ (86.85 and 85.74, respectively), indicating that a

moderate neighborhood size effectively aggregates essential topological context. However, performance declines beyond $k = 7$. This is likely due to the over-smoothing effect, where excessively large neighborhoods dilute fine-grained vascular features with redundant structural noise. Additionally, computational costs increase monotonically with k. Given the diminishing returns and escalating overhead, we selected $k = 7$ as the optimal trade-off between topological awareness and inference efficiency.

5 Conclusion

We presented GTUNet, a topology-aware hybrid framework for 3D coronary artery segmentation. By integrating 3D ResUNet, Swin Transformer, and Graph Attention Networks (GAT), GTUNet bridges local texture perception, long-range contextual modeling, and non-Euclidean structural priors. Its key components, including the GAT-based enhancement module and the Bidirectional Attention Fusion Module (BAFM), promote effective information exchange and incorporate structural connectivity into semantic feature learning.

Experiments on the ASOCA and ImageCAS datasets demonstrate that GTUNet achieves state-of-the-art performance compared with four representative methods, while effectively reducing disconnection artifacts in distal branches and complex bifurcations. The improved topological fidelity may provide a reliable basis for downstream clinical applications, such as anatomical modeling and FFR-CT functional assessment. Future work will investigate the generalization of this framework to other vascular networks, including cerebral and pulmonary vessels.

Author Contributions Mengjun Sheng wrote the main manuscript text. All authors performed the experiments, analyzed the data, and reviewed the manuscript.

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Data Availability The datasets analyzed in this study are publicly available.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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